

ROHIT SINGH

AI/ML Engineer • Computer Vision • Geospatial Intelligence • Foundation Models

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🌐 Portfolio / Demos 📍 Ghaziabad, UP, India

EDUCATION

ABES Engineering College (Affiliated to AKTU, Lucknow)

Aug 2025 – Expected July 2029

Bachelor of Technology (B.Tech) in Computer Science & Engineering (AI & ML)

Cumulative CGPA: 8.8 / 10.0 (88.0%) • 1st Year YGPA: 8.79 (46/46 Credits)

Relevant Coursework: Data Structures & Algorithms in C++, Object-Oriented Programming, Database Management Systems, Linear Algebra, Probability & Statistics, Machine Learning Foundations, Discrete Mathematics.

TECHNICAL SKILLS

Languages: Python 3.11, C++ (Advanced DSA), JavaScript (ES6+), TypeScript, SQL (PostgreSQL), Bash/Shell.

Machine Learning & Vision: PyTorch, OpenCV (Perceptual Image Hashing dHash), Hugging Face Transformers, LangChain, Scikit-Learn, NetworkX (Graph Theory).

Geospatial & Remote Sensing: Landsat-8/9 TIRS (Radiometric Calibration & Planck Inversion), Sentinel-2 MSI (NDVI & FVC), GeoPandas, Rasterio, Leaflet.js.

Systems & Deployment: FastAPI, Docker (Multi-Stage Builds), ChromaDB (Vector Store), React 18, Vite, Git/GitHub, Linux/UNIX.

FLAGSHIP ENGINEERING & RESEARCH PROJECTS

HeatLens: Autonomous Satellite Geospatial & Urban Climate AI Oracle

[Live Vercel Portal] [GitHub Code]

- Engineered an autonomous Earth observation pipeline ingesting multispectral satellite rasters from **Landsat-8/9 TIRS (Band 10, 30m)** and **Sentinel-2 MSI (NDVI, 10m)** to analyze localized microclimate disparities across informal urban settlements.
- Calibrated raw 16-bit Digital Numbers to spectral radiance and inverted **Planck's Radiation Law** to compute Land Surface Temperature (LST); computed Fractional Vegetation Cover (FVC) to derive ground surface emissivity.
- Formulated a composite priority index: $P = 0.35(LST) + 0.25(1-NDVI) + 0.20(AQI) + 0.20(Risk)$ to simulate NDMA Heat Action Plan cooling interventions. Developed for **Build With Bharat 2.0 (NIT Delhi)** with **x402 Algorand** oracle verification.

MPLADS-AI: Civic Infrastructure Computer Vision Forensics & Cartel Anomaly Engine

[Live Vercel Portal] [Team Repo]

- Architected an end-to-end continuous surveillance layer for the **Ministry of Statistics & Programme Implementation (MoSPI / e-SAKSHI)** auditing public works proposals across 700+ administrative districts.
- Implemented **64-bit Difference Hashing (OpenCV dHash)** with Hamming distance matching to identify recycled and fraudulent site photos with **96% precision**, featuring a live bitstream differential analyzer.
- Modeled procurement bidding relationships as **Bipartite Graphs via NetworkX**; computed **Herfindahl-Hirschman Index (HHI)** to flag suspicious tender concentration and contractor cartels. Shortlisted in **Top 20 out of 416 teams** at Smart India Hackathon (SIH 2026).

World Data AI: Agentic Retrieval-Augmented Generation (RAG) & Knowledge Engine

[GitHub Code]

- Architected an enterprise Agentic RAG pipeline using **LangChain Expression Language (LCEL)** and **LLaMA-3** for multi-hop semantic querying over complex global datasets.
- Engineered recursive character chunking and dense vector embeddings with **ChromaDB**, integrating ReAct tool-calling loops for evidence synthesis and hallucination-free document reasoning.

EXPERIENCE & ACADEMIC LEADERSHIP

Undergraduate AI & Computer Vision Researcher • ABES Engineering College

Sep 2025 – Present

- Leading architectural design and algorithmic development for civic forensics and satellite climate analytics pipelines.
- Mentored under faculty in the Department of Computer Science & Engineering (AI & ML) on deep learning optimization and spatial statistics.

Technical Team Lead • Code Titans (Competitive Programming & Hackathons)

Jan 2026 – Present

- Directing high-stakes competitive hackathon builds; mentoring peers in C++ Data Structures, Algorithm design, and applied AI systems.

VERIFIED CREDENTIALS & NATIONAL TRAINING

DataCamp: Working with Hugging Face Transformers ID: 49500149

DataCamp: Developing LLM Apps with LangChain ID: 49395951

CodeChef PRO: DSA in C++ (Advanced Algorithms) Cert ID: 9ccb774

IBM SkillsBuild: AI & Machine Learning Foundations PLAN-E624C2604060

ISRO (IIRS): Climate Change-Induced Disasters & Space Remote Sensing Training Programme

National Space Agency